Lecture 4 Linear Regression II

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

Multiple Linear Regression

- Multiple Linear Regression has more than one covariates,

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon,$$

  where usually $\varepsilon \sim N(0, \sigma^2)$.
- We interpret β_j as the average effect on Y of a one unit increase in X_j, while holding all the other covariates fixed.
- In the advertising example, the model becomes

$$\text{Sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{Radio} + \beta_3 \times \text{Newspaper} + \varepsilon.$$

Coefficient Interpretation

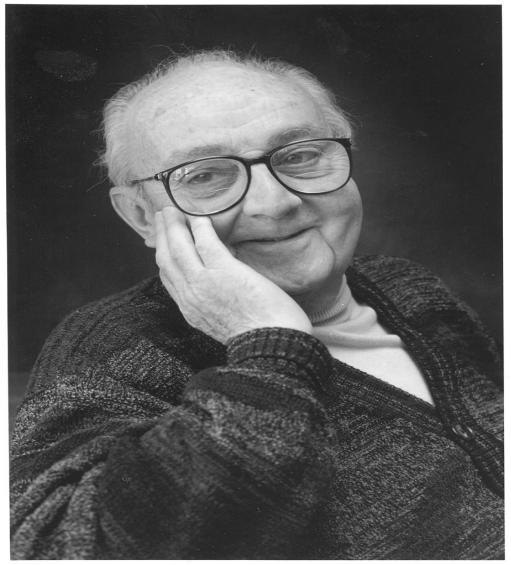
- The ideal scenario is when the predictors are uncorrelated — a balanced design.
  - Each coefficient can be estimated and tested separately.
  - Interpretations such as a unit change in X_j is associated with a β_j change in Y , while all the other variables stay fixed, are possible.
- Correlations amongst predictors cause problems.
  - The variance of all coefficient tends to increase, sometimes dramatically.
  - Interpretations become hazardous — when X_j changes, everything else changes.
- Claims of causality should be avoided for observational data.

The woes of regression coefficients

Data Analysis and Regression, Mosteller and Tukey 1977

- A regression coefficient β_j estimates the expected change in Y per unit change in X_j, with all other predictors held fixed. But predictors usually change together!
- Example: Y total amount of change in your pocket; $X_1 = \#$ of coins; $X_2 = \#$ of pennies, nickels and dimes. By itself, regression coefficient of Y on X_2 will be > 0. But how about with X_1 in model?
- $Y =$ number of tackles by a football player in a season; W and H are his weight and height. Fitted regression model is $Y = \beta_0 + 0.50W - 0.10H$. How do we interpret $\hat{\beta}_2 < 0$?

Two quotes by famous Statisticians



1919 - 2013 (aged 93)

- Essentially, all models are wrong, but some are useful.
  George Box
- The only way to find out what will happen when a complex system is disturbed is to disturb the system, not merely to observe it passively.
  Fred Mosteller and John Tukey, paraphrasing George Box

Coefficient estimation

- Given the estimates $\hat{\beta}_0, \hat{\beta}_1, \cdots$, and $\hat{\beta}_p$, the estimated regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p.$$

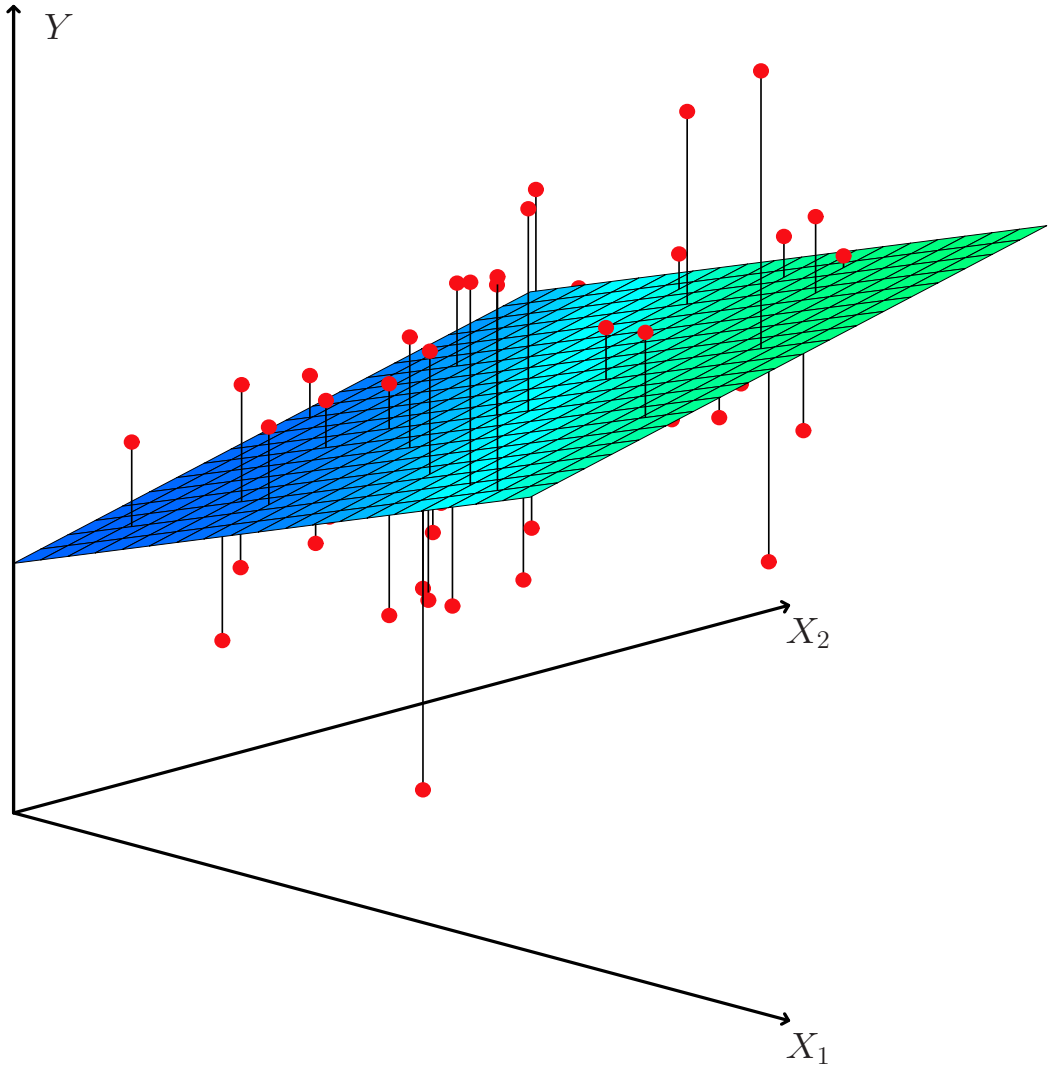
- We estimate all the coefficients $\beta_j, j = 0, 1, \cdots, p$ as the values that minimize the sum of squared residuals

$$\text{RSS} = \sum_i (y_i - \hat{y}_i)^2,$$

where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}$ is the predicted values.

- This is done using standard statistical software. The values $\hat{\beta}_0, \hat{\beta}_1, \cdots$, and $\hat{\beta}_p$ that minimize RSS are the multiple least squares regression coefficient estimates.

Estimation Example



Number of rows

Observations: y_i, x_{ij}, for $i = 1, \ldots, n$ and $j = 0, 1, \ldots, p$. Noise: $\varepsilon_1, \ldots, \varepsilon_n$.
Unknown parameters: $\beta_0, \beta_1, \ldots, \beta_p$, which we try to mine data for.

$$
\begin{aligned}
y_1 &= \beta_0 + \beta_1 x_{11} + \cdots + \beta_p x_{1p} + \varepsilon_1 \\
y_2 &= \beta_0 + \beta_1 x_{21} + \cdots + \beta_p x_{2p} + \varepsilon_2 \\
&\vdots \\
y_n &= \beta_0 + \beta_1 x_{n1} + \cdots + \beta_p x_{np} + \varepsilon_n
\end{aligned}
$$

We write

$$\mathbf{Y} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \boldsymbol{\varepsilon},$$

or,

$$\underset{(n\times 1)}{\mathbf{Y}} = \underset{(n\times(p+1))}{\mathbf{X}} \underset{((p+1)\times 1)}{\boldsymbol{\beta}} + \underset{n\times 1}{\boldsymbol{\varepsilon}},$$

which is an elegant and equivalent matrix form.

Least squares estimation

$$
\begin{aligned}
RSS &= \sum_{j=1}^{n} (y_j - b_0 - b_1 x_{j1} - \cdots - b_p x_{jp})^2 \\
&= (\mathbf{y} - \mathbf{X}\mathbf{b})'(\mathbf{y} - \mathbf{X}\mathbf{b})
\end{aligned}
$$

When $\mathbf{X}$ has full rank $p + 1 \leq n$, the least square estimate of β is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$ → ML starts from here.

Derivation:

- Let $x \in \mathbb{R}^m$, A, B be constant matrices, and c be a constant vector. ?

$$D_x \langle Ax, c \rangle = A'c, \quad D_x \langle Ax, Bx \rangle = B'Ax + A'Bx.$$

- Therefore $D_b RSS = D_b \langle Xb - y, Xb - y \rangle = 2X'(Xb - y)$.
- We let $D_b RSS = 0$ to obtain the estimate.

Example

Fit a straight-line regression model

$$Y = \beta_0 + \beta_1 X_1 + \varepsilon,$$

with data

X_1	0	1	2	3	4
Y	1	4	3	8	9

.

Solution.

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 4 \\ 3 \\ 8 \\ 9 \end{bmatrix}, \quad X'X = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}.$$

Recall that if

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

and if A is invertible, then $|A| = a_{11}a_{22} - a_{12}a_{21}$ and

$$\frac{1}{|A|}\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

So

$$A^{-1} = \frac{1}{|A|}\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}, \text{ and}$$

$$(X'X)^{-1} = \frac{1}{5 \times 30 - 10 \times 10}\begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix}.$$

On the other hand,

$$X'Y = \begin{bmatrix} 25 \\ 70 \end{bmatrix},$$

so

$$\hat{\beta} = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix} \begin{bmatrix} 25 \\ 70 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Inference

- Is at least one predictor useful?
  - $H_0 : \beta_1 = \cdots = \beta_p = 0, \qquad \text{v.s.} \qquad H_1 : \text{otherwise}$
  - Insight: to test if there is a linear relationship between Y and $X_1, \ldots, X_p$.
- Test statistic $\mathrm{F} = \frac{(\mathrm{TSS}-\mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)} \sim F_{p,n-p-1}$.
- $$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2, \quad TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$$
- Under H_0, $F \sim F_{p,n-p-1}$, the F distribution with degrees of freedom p and $n-p-1$. One rejects H_0 if F is too large (depending on the test significance), in which case $\hat{\beta}$ is significant.

Inference

Only calculated (t-test) when Ho is rejected.

- What about an individual coefficient, say if β_j useful?
- $H_0 : \beta_j = 0, \qquad \text{v.s.} \qquad H_1 : \beta_j \neq 0.$
- Test statistic: $t = \frac{\hat{\beta}_j - 0}{\mathrm{SE}(\hat{\beta}_j)} \sim t_{n-p-1}$.
- t_{n-p-1}, the t-distribution with $n - p - 1$ degrees of freedom.
- $\mathrm{SE}(\hat{\beta}_j)$ is called the **estimated standard error**, or simply **standard error** of the coefficient $\hat{\beta}_j$,

$$\mathrm{SE}(\hat{\beta}_j) = \sqrt{\frac{\mathrm{RSS}}{n-p-1}\left[(X'X)^{-1}\right]_{jj}},$$

where $\left[(X'X)^{-1}\right]_{jj}$ is the (j,j)-entry of the matrix $(X'X)^{-1}$.

R^2 and Adjusted R^2

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- $R^2 \in [0, 1]$ is called the **coefficient of determination**. It gives the proportion of the total variation in the y_j's, that is "explained" by, or attributable to, the predictor variables $X_1, \ldots, X_p$.
- $R^2 = 1$ if the fitted equation passes through all the data point (perfect fit, $\hat{y}_j = y_j$ for any j).
- $R^2 = 0$ if $\hat{\beta}_0 = \bar{y}$ and $\hat{\beta}_1 = \cdots = \hat{\beta}_p = 0$, in which case $X_1, \ldots, X_p$ have no influence on the response Y.

R^2 and Adjusted R^2

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- The adjusted R^2 statistic

$$R^2_{\text{Adj}} = 1 - \frac{\text{RSS}/(n-r-1)}{\text{TSS}/(n-1)}.$$

- since $\text{TSS} = \text{RSS} + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$ (we skip the proof),

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

- In general, R^2 never decreases when one more prediction variable (e.g. X_{p+1}) is added to the model, even if the new predictor contributes nothing.
- $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2/(n-r-1)$ is the residual mean square. When one more predictor is added in, R^2_{Adj} is increased only if $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2/(n-r-1)$ is decreased.

Estimating the regression function at x_0

For the linear regression model $Y = X\beta + \varepsilon$, $x_0'\hat{\beta} = \hat{\beta}_0 + \hat{\beta}_1 x_{01} + \cdots + \hat{\beta}_p x_{0p}$ is the unbiased linear estimator of $\mathbb{E}[Y|x_0]$. If the errors $\varepsilon_1, \ldots, \varepsilon_n$ are independently distributed and all have $N(0, \sigma^2)$ distribution then a $100(1-\alpha)\%$ confidence interval for $\mathbb{E}[Y|z_0] = z_0'\beta$ is given by

$$x_0'\hat{\beta} \pm t_{n-p-1}\left(\frac{\alpha}{2}\right)\sqrt{(x_0'(X'X)^{-1}x_0)s^2},$$

where $s^2 = \frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.

Advertising example

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.938889   0.311908   9.422   <2e-16 ***
TV           0.045765   0.001395  32.809   <2e-16 ***
Radio        0.188530   0.008611  21.893   <2e-16 ***
Newspaper   -0.001037   0.005871  -0.177     0.86
---
Signif. codes:  0 ?***? 0.001 ?**? 0.01 ?*? 0.05 ?.? 0.1 ? ? 1

Residual standard error: 1.686 on 196 degrees of freedom
Multiple R-squared:  0.8972,Adjusted R-squared:  0.8956
F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16

> predict(TVadlm, newdata, interval="c", level=0.95)
fit      lwr      upr
1 20.52397 19.99627 21.05168
> predict(TVadlm, newdata, interval="p", level=0.95)
fit      lwr      upr
1 20.52397 17.15828 23.88967
```

Confidence interval

Attributes of Data

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There are different attributes/types of data.

- **nominal**: being something in name only, and not in reality. Examples: ID numbers, eye color, postal codes Only to identify
- **Ordinal**: referring to or connected with the position of something in a series. Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short} have an order.
- **Binary**: a special kind of nominal attribute which only has two categories/states: 0 or 1. Sometimes referred to as Boolean if the two states are called `true` and `false`. Donot represent ordinality
- **Interval-Scaled**: measured on a scale of equal-size units. Examples: date, temperature. No "true zero", so we can NOT say $10°C$ is twice as warm as $5°C$. We still can compute mean and median.
- **Ratio-Scaled**: numeric attribute with an inherent zero-point. Examples: height, weight, distance.
- many other types....

Data quality

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- What kinds of data quality problems? → cleaned before data analysis.
- How can we detect problems with the data?
- What can we do about these problems?
- Examples of data quality problems:
  - Noise and outliers
  - missing values
  - duplicate data

Indicator Variables

- Some predictors are not quantitative but are qualitative, taking a discrete set of values.
- These are also called categorical predictors or factor variables.
- Example: investigate difference in credit card balance between males and females, ignoring the other variables. We create a new variable,

$$x_i = \begin{cases} 1 & \text{if } i\text{-th person is female,} \\ 0 & \text{if } i\text{-th person is male} \end{cases}.$$

- Resulting model

1 -> female

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{-th person is female,} \\ \beta_0 + \varepsilon_i & \text{if } i\text{-th person is male} \end{cases}.$$

- Interpretation and more than two levels (categories)?

Indicator Variables

- In general, if we have k levels, we need $(k-1)$ indicator variables.
- For example, we have 3 levels — $A, B,$ and C for a covariate x,

$$x_A = \begin{cases} 1 & \text{if } x \text{ is A,} \\ 0 & \text{if } x \text{ is not A} \end{cases}; \quad x_B = \begin{cases} 1 & \text{if } x \text{ is B,} \\ 0 & \text{if } x \text{ is not B} \end{cases}.$$

- If x is C, then $x_A = x_B = 0$. We call C as baseline.
- β_A is the contrast between A and C and β_B is the contrast between B and C.

Missing values

- Reasons for missing values
  - Information is not collected (e.g., people decline to give their age and weight)
  - Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)
- Handling missing values
  - Eliminate Data Objects
  - Estimate Missing Values
  - Ignore the Missing Value During Analysis
  - Replace with all possible values (weighted by their probabilities)

Duplicate data

- Data set may include data objects that are duplicates, or almost duplicates of one another (Major issue when merging data from heterogeneous sources)
- Examples: Same person with multiple email addresses
- Data cleaning Process of dealing with duplicate data issues

Summary and Remark

- Multiple linear regression
- Estimation and inference
- Data types
- Read textbook Chapter 3